

Progressive covariance compression, direct Cartan Fourier trace, and rational mean-spectral boundary

TECT verification-first repository

claim A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION

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1 Purpose and scope

R-088 left two central analytic questions. First, its quartic terminal bridge was stated only for a regular orthogonal one-shot control class. Second, its production Cartan estimate was organized atom by atom even though the exact root expression permits all physical control shells to be summed before the Hilbert–Schmidt square. The rational packet also retained a same-root conditional defect whose sign had not been tested on the actual production scalar ray.

This note proves four results and records two exact method no-gos.

1. A global polar/Douglas contraction controls the terminal Cameron–Martin norm of every finite-cutoff cylindrical-simple progressive control, independently of time partition, range overlap, or revisit multiplicity. Hilbert-valued martingale orthogonality then gives a genuine one-use terminal-shell ledger. Consequently the R-088 pure-control quartic terminal bridge extends to the full progressive class.
2. The complete sum over physical control shells in the Cartan derivative is one first-order operator. An exact Fourier trace identity, retaining all cross-shell terms, reduces its far output to two nonlinear coefficient tails. This avoids both output orthogonality and a per-atom ledger.
3. The rational conditional packet has a simpler Taylor-coordinate normal form. Under covariance matching it is universally nonnegative exactly when two displayed matrices are positive semidefinite. The actual production scalar ray has a negative mean matrix, and a centered, covariance-matched, same-root adapted fixture has strictly negative expected packet.
4. The R-087 variational CORE is exactly equivalent to the uniform $q = 10/9$ Nelson estimate. Therefore full OVERLAP must already contain the complete temporal packet and controlled-shell assembly; invoking those ingredients again after CORE is logically redundant.

The note does not prove the nonlinear Cartan coefficient-tail energy, the complete same-root rational heat/forest packet, full REG, uniform OVERLAP, the Nelson estimate, an interacting measure, or Sector-A closure. Tier stays T4.

The load-bearing internal authorities are A8 and R-076, R-081, R-084, R-087, and R-088. The A8 covariance symbol and regulator bounds are used only for the Cameron–Martin to H^2 comparison. R-087 supplies the exact root-diagonal Cartan target and variational CORE. R-088 supplies the sequential secant and quartic Besov endpoint.

2 Global progressive covariance compression

Fix a finite cutoff J . Let $\mathcal{U} = L^2([0, 1]; H_0)$ be the control space, let H_J be the finite-dimensional terminal Gaussian space, and let

$$T_J v = \int_0^1 K_J(t) v_t dt, \quad C_J = T_J T_J^* = \int_0^1 K_J(t) K_J(t)^* dt. \quad (2.1)$$

No disjoint-range or one-shot assumption is imposed on $K_J(t)$.

Theorem 2.1 (pathwise global contraction)

For every $v \in \mathcal{U}$,

$$\|C_J^{\dagger/2} T_J v\|_{H_J}^2 \leq \int_0^1 \|v_t\|_{H_0}^2 dt. \quad (2.2)$$

Here $\dagger$ is the Moore–Penrose inverse, so the left side is the canonical Cameron–Martin norm on $\text{Ran } C_J^{1/2}$.

Proof. The polar decomposition of T_J can be written

$$T_J = C_J^{1/2} R_J, \quad (2.3)$$

where $R_J : \mathcal{U} \rightarrow H_J$ is a contraction whose range lies in $\overline{\text{Ran } C_J}$. Thus $C_J^{\dagger/2} T_J = P_{\text{Ran } C_J} R_J$ and has operator norm at most one. Squaring proves (2.2). $\square$

The theorem is deterministic and pathwise. It therefore applies to a random progressive v after fixing the sample. In particular, revisiting the same terminal direction many times cannot create a multiplicity loss: the same overlap appears in both $T_J v$ and $C_J = T_J T_J^*$ and is normalized by the pseudoinverse.

Under the A8 production symbol bound

$$\mathcal{A}(k) \succeq c_{\text{sym}} \langle k \rangle^4 I \quad (2.4)$$

and the regulator multiplier bound $\|R_J\| \leq M_R$, (2.2) yields

$$\|T_J v\|_{H^2}^2 \leq \frac{M_R^2}{c_{\text{sym}}} \int_0^1 \|v_t\|^2 dt. \quad (2.5)$$

This is the finite-cutoff Cameron–Martin embedding with constants uniform in J .

Hilbert martingale and spatial ledgers

Let $\mathcal{F}_j$ reveal the Gaussian roots in increasing order and put

$$d_j A = \mathbb{E}[A \mid \mathcal{F}_j] - \mathbb{E}[A \mid \mathcal{F}_{j-1}], \quad A = T_J v. \quad (2.6)$$

Martingale differences are orthogonal in the Cameron–Martin Hilbert space, hence

$$\sum_j \mathbb{E} \|d_j A\|_{\text{CM}}^2 \leq \mathbb{E} \|A\|_{\text{CM}}^2 \leq \mathbb{E} \int_0^1 \|v_t\|^2 dt. \quad (2.7)$$

The same argument after the deterministic H^2 multiplier in (2.5) gives

$$\sum_j \mathbb{E} \|d_j A\|_{H^2}^2 \leq \frac{M_R^2}{c_{\text{sym}}} \mathbb{E} \int_0^1 \|v_t\|^2 dt. \quad (2.8)$$

For an inhomogeneous Littlewood–Paley decomposition, every $0 < s \leq 2$, and roots $j \geq j_0$,

$$\boxed{\sum_{j,m} 2^{2s(m-j)} \mathbb{E} \|\Delta_m d_j A\|_2^2 \leq C_{s,j_0} \frac{M_R^2}{c_{\text{sym}}} \mathbb{E} \int_0^1 \|v_t\|^2 dt.} \quad (2.9)$$

Indeed, $2^{-2sj} \|d_j A\|_{H^s}^2$ is bounded by a constant times $\|d_j A\|_{H^2}^2$, and (2.8) sums the latter exactly once. Restricting (2.9) to $m \geq j + C$ gives the far gain

$$\sum_j \sum_{m \geq j+C} \mathbb{E} \|\Delta_m d_j A\|_2^2 \leq C_{s,j_0} 2^{-2sC} \frac{M_R^2}{c_{\text{sym}}} \mathbb{E} \int_0^1 \|v_t\|^2 dt. \quad (2.10)$$

General progressive quartic terminal bridge

R-088 proved, for $0 < s < 1$,

$$\|A^{\otimes 3} \otimes DA\|_{B_{1,1}^s} \lesssim_s \|A\|_{H^2} \|A\|_6^3. \quad (2.11)$$

Equation (2.5) removes its former one-shot restriction. If $X = \int_0^1 \|v_t\|^2 dt$ and $Y = \|U + A\|_6^6$, then

$$\mathbb{E}[\|A\|_{H^2} \|A\|_6^3] \leq C \mathbb{E}[X^{1/2} Y^{1/2}] + \delta \mathbb{E}X + C_\delta \mathbb{E}\|U\|_6^6. \quad (2.12)$$

The final model moment is uniform by the accepted Gaussian input. Thus the pure-control quartic terminal payload is form-payable for arbitrary finite-cutoff cylindrical-simple progressive controls. This conclusion does not estimate the lower-order random coefficient tails in the full Cartan atom.

3 Direct integrated Cartan Fourier trace

Fix a generator A and root j . Sum the R-088 sequential physical shells before taking any square. Write

$$Z_j^- = w_j + A^{(\ell)}, \quad a^{[j]} = A^{(j)} - A^{(\ell)}, \quad Z_j^+ = Z_j^- + a^{[j]}. \quad (3.1)$$

For the production map $\Phi = \Phi_{A,j}$ define

$$c_{A,j} = \Phi(Z_j^+) - \Phi(Z_j^-) \quad (3.2)$$

and, for each spatial coordinate i ,

$$\begin{aligned} b_{A,j,i} &= [D\Phi(Z_j^+) - D\Phi(Z_j^-)]^T \partial_i Z_j^- \\ &\quad + D\Phi(Z_j^+)^T \partial_i a^{[j]}. \end{aligned} \quad (3.3)$$

The exact telescope and the product rule give

$$\boxed{\sum_{k \leq j} T_{A,j,k}[v] = L_{A,j} v, \quad (L_{A,j} v)_i = b_{A,j,i}^T v + c_{A,j}^T \partial_i v.} \quad (3.4)$$

This identity contains every cross term between different physical control shells. Ornstein–Uhlenbeck averaging preserves the first-order form; below $b_{A,j,t}$ and $c_{A,j,t}$ denote the averaged coefficients.

Let the root covariance have Fourier eigenvectors

$$v_{p,\mu}(x) = \sqrt{\lambda_{j,p}} e^{ipx} e_\mu, \quad p \in S_j \subset Q_j, \quad (3.5)$$

and suppose

$$\Lambda_{0,j} := \sum_{p,\mu} \lambda_{j,p} \leq \kappa_0 2^{-j}, \quad \Lambda_{1,j} := \sum_{p,\mu} \lambda_{j,p} |p|^2 \leq \kappa_1 2^j. \quad (3.6)$$

Theorem 3.1 (exact trace and two-tail bound)

For an output multiplier Π_m and $m \geq j + 3$,

$$\begin{aligned} \|\Pi_m L_{A,j,t} \Gamma_j^{1/2}\|_{\text{HS}}^2 &= \sum_{\mu,p} \lambda_{j,p} \sum_q |\pi_m(q)|^2 \\ &\quad \times \sum_i |\widehat{b}_{A,j,t,i}(q-p)e_\mu + ip_i \widehat{c}_{A,j,t}(q-p)e_\mu|^2. \end{aligned} \quad (3.7)$$

If $\widetilde{\Pi}_m$ selects $2^{m-2} < |r|_\infty \leq 2^{m+2}$, then

$$\|\Pi_m L_{A,j,t} \Gamma_j^{1/2}\|_{\text{HS}}^2 \leq 2\Lambda_{0,j} \|\widetilde{\Pi}_m b_{A,j,t}\|_2^2 + 2\Lambda_{1,j} \|\widetilde{\Pi}_m c_{A,j,t}\|_2^2. \quad (3.8)$$

Proof. Apply (3.4) to each vector (3.5). The Fourier coefficient at output q is exactly the vector displayed in (3.7), so Parseval and summation over the root basis give equality. When q is in shell m , p is in shell j , and $m \geq j + 3$, the difference $q - p$ lies in the enlarged annulus. Apply $|x + y|^2 \leq 2|x|^2 + 2|y|^2$, sum first in q , and then use (3.6). This proves (3.8). $\square$

The enlarged annuli have overlap at most five. Summing (3.8) over $m \geq j + C$ and restoring the R-084 prefactor $3/(40P)$ yields

$$S_C^{\text{ctrl}} \leq \frac{3}{4P} \sum_{A,j} \int_0^\infty e^{-2t} \mathbb{E} \left[\kappa_0 2^{-j} \|Q_{j,C-2}^{\text{coef}} b_{A,j,t}\|_2^2 + \kappa_1 2^j \|Q_{j,C-2}^{\text{coef}} c_{A,j,t}\|_2^2 \right] dt. \quad (3.9)$$

No output orthogonality and no per- k quantity q_k was used.

For $s > 0$, Sobolev tail extraction reduces (3.9) to

$$S_C^{\text{ctrl}} \leq \frac{3}{4P} 2^{-2s(C-2)} \mathfrak{E}_s, \quad (3.10)$$

where

$$\mathfrak{E}_s = \sum_{A,j} \int_0^\infty e^{-2t} \mathbb{E} \left[\kappa_0 2^{-(2s+1)j} \|b_{A,j,t}\|_{H^s}^2 + \kappa_1 2^{(1-2s)j} \|c_{A,j,t}\|_{H^s}^2 \right] dt. \quad (3.11)$$

The remaining Cartan theorem is exactly

$$\mathfrak{E}_s \lesssim 1 + \mathbb{E}[X^{1/2} Y^{1/2}] \quad (3.12)$$

for some $s > 0$. Equations (3.7)–(3.11) prove the direct integrated reduction, not (3.12).

4 A subcritical quartic route and its exact boundary

The R-076 interpolation proof gives the stronger family

$$\|A^{\otimes 3} \otimes DA\|_{B_{1,1}^s} \lesssim_s \|A\|_{H^2}^{(1+s)/2} \|A\|_6^{(7-s)/2}, \quad 0 < s < 1. \quad (4.1)$$

At $s = 1/4$, in terms of $X_A = \|A\|_{H^2}^2$ and $Y_A = \|A\|_6^6$, its powers are

$$X_A^{5/16} Y_A^{9/16}. \quad (4.2)$$

There is Young slack $1/8$, so a model coefficient requires only an eighth moment; the corresponding small-parameter losses are $\eta^{-5/2}$ and $\zeta^{-9/2}$. The direct Schur constant and gap are

$$\frac{1}{(1 - 2^{-1/4})^2 (1 - 2^{-1/2})} = 134.874781327179 \dots, \quad 2^{-C/2}. \quad (4.3)$$

This subcritical ledger has genuine slack and is a plausible route for the quartic part of (3.12). It does not control the lower-order b, c terms.

Exact no-go for a pure homogeneous quartic bridge

The missing lower-order terms cannot be deleted. On the normalized scalar production ray let

$$F(x) = \frac{x^3}{1+x^2}, \quad B(x) = \cos x, \quad a_c(x) = c \cos x, \quad v(x) = \cos x. \quad (4.4)$$

The exact sequential atom is

$$T_c[v] = \frac{d}{dx} \{ [F(B + a_c) - F(B)]v \} = cH'(x) + O(c^2), \quad (4.5)$$

where

$$H(x) = F'(\cos x) \cos^2 x = \frac{\cos^4 x (3 + \cos^2 x)}{(1 + \cos^2 x)^2}. \quad (4.6)$$

Its cosine harmonic at frequency 32 is exactly

$$h_{32} = \frac{\sqrt{2}}{2} (102\sqrt{2} - 137)(3 - 2\sqrt{2})^{15} = 1.684765541129627 \times 10^{-11} > 0. \quad (4.7)$$

Hence H' has sine coefficient $-32h_{32}$ and normalized shell energy

$$\frac{(32h_{32})^2}{2} = 1.453278683431837 \times 10^{-19} > 0. \quad (4.8)$$

The squared atom therefore scales as c^2 , while $\|a_c^{\otimes 3} Da_c\|$ and $X_{a_c}^{1/2} Y_{a_c}^{1/2}$ scale as c^4 . No uniform homogeneous bound using only that quartic payload can imply the complete production far atom. The statement fires [NG-2026-07-25-A13-PURE-QUARTIC-CARTAN-HOMOGENEITY](#). It does not refute a form bound with additive constants, model moments, or the lower-order background/control coefficient tails retained. The positive floor transfers by $F_e(\sqrt{e}, y) = \sqrt{e}, F_1(y)$.

5 Rational Taylor-coordinate normal form

Use the R-088 notation

$$Q = GG^T - \Gamma, \quad B_1 = B_T + L, \quad (5.1)$$

and

$$\mathcal{P} = \frac{1}{2}L : Q + G^T Lc + \frac{1}{2}c^T B_1 c. \quad (5.2)$$

Let $\mathcal{H}$ contain the coefficients and put $\mu = \mathbb{E}[G \mid \mathcal{H}]$ and $V = \text{Cov}(G \mid \mathcal{H})$. Direct conditional expectation and completion in Taylor coordinates give the exact identity

$$\boxed{\mathbb{E}[\mathcal{P} + \eta|c|^2 \mid \mathcal{H}] = \frac{1}{2}(c + \mu)^T L(c + \mu) + \frac{1}{2}c^T (B_T + 2\eta I)c + \frac{1}{2}L : (V - \Gamma).} \quad (5.3)$$

This is algebraically equivalent to the R-088 Feshbach form but exposes the two independent spectral channels.

Theorem 5.1 (mean-spectral criterion)

Assume $V = \Gamma$. The right side of (5.3) is nonnegative for every c, μ if and only if

$$\boxed{L \succeq 0 \quad \text{and} \quad B_T + 2\eta I \succeq 0.} \quad (5.4)$$

Proof. Put $x = c + \mu$. The pair (x, c) ranges independently over the product space, and (5.3) becomes $x^T L x / 2 + c^T (B_T + 2\eta I) c / 2$. Nonnegativity for all pairs is therefore equivalent to the two matrix inequalities. $\square$

In particular, increasing η cannot repair a negative direction of L : choose $c = 0$ and put the conditional mean in that direction.

Production scalar obstruction

Remove the harmless positive rational prefactor and scale the floor to one. For

$$f(u) = u - \frac{5}{9} \frac{u^3}{1 + u^2}, \quad B(u) = f(u)^2, \quad (5.5)$$

take base point $z = 1$ and increment $a = C$. The endpoint-minus-quadratic Taylor remainder is

$$L_{\text{norm}}(C) = \frac{5C^3(13C^3 + 36C^2 + 51C + 20)}{324(C^2 + 2C + 2)^2}. \quad (5.6)$$

At $C = -1/2$,

$$\frac{B_1}{e} = \frac{16}{81}, \quad \frac{B_T}{e} = \frac{259}{1296}, \quad \frac{L}{e} = -\frac{1}{432}. \quad (5.7)$$

Restoring $B_R = 4c_1 B$, choosing $c = 0$, $V = \Gamma$, and a scalar mean $\mu = t$ makes (5.3)

$$-\frac{c_1 e}{216} t^2 < 0, \quad (5.8)$$

for every η . This is a production-coefficient method no-go, not a counterexample to the full heat/forest form.

There is also a centered covariance-matched same-root fixture. Let $G \sim N(0, 1)$, $Q = G^2 - 1$, $c = 0$, and choose

$$C(G) = \begin{cases} -1/2, & Q \geq 0, \\ +1/2, & Q < 0. \end{cases} \quad (5.9)$$

The other Taylor remainder is $L_{\text{norm}}(1/2) = 2245/219024$. Since $\mathbb{E}[Q \mathbf{1}_{\{Q \geq 0\}}] = 2\varphi(1)$,

$$\boxed{\mathbb{E}\mathcal{P} = -\frac{688}{13689} c_1 e \varphi(1) < 0.} \quad (5.10)$$

Smooth bounded even cylindrical approximations to (5.9) preserve the sign by dominated convergence. Thus fresh-root centering, covariance matching, Jensen, and the completion parameter alone cannot close the same-root production packet. This fires [NG-2026-07-25-A13-RATIONAL-ETA-MEAN-SPECTRAL-CLOSURE](#).

If $\mathcal{F}$ denotes the complete backward-heat and lower-chaos forest, define the exact residual

$$\mathcal{R}_{\mathcal{H}} := \mathbb{E}[\mathcal{F} \mid \mathcal{H}] + \mu^T L c + \frac{1}{2} \mu^T L \mu + \frac{1}{2} L : (V - \Gamma). \quad (5.11)$$

Then

$$\boxed{\mathbb{E}[\mathcal{P} + \eta |c|^2 + \mathcal{F} \mid \mathcal{H}] = \frac{1}{2} c^T (B_1 + 2\eta I) c + \mathcal{R}_{\mathcal{H}}.} \quad (5.12)$$

The remaining rational theorem is a production spatial/causal lower bound on $\mathcal{R}_{\mathcal{H}}$ with the endpoint square retained. The fixture (5.10) shows why the forest cannot be omitted.

6 OVERLAP–Nelson equivalence and corrected order

Put

$$G_J = V_J^{\text{ren}} + \frac{3}{20} \|\cdot\|_6^6, \quad I_J(v) = \mathbb{E} G_J(X_J + T_J v) + \frac{9}{20} \mathbb{E} \int_0^1 |v_t|^2 dt. \quad (6.1)$$

The R-087 CORE identity is

$$\boxed{\inf_{v \in \mathcal{A}_{\text{cs}}} I_J(v) = -\frac{9}{10} \log \mathbb{E} \exp\{-\frac{10}{9} G_J(X_J)\}.} \quad (6.2)$$

Consequently, for the same constant C ,

$$I_J(v) \geq -C \quad \text{for every } J, v \quad (6.3)$$

is equivalent to

$$\sup_J \mathbb{E} \exp\{-\frac{10}{9} G_J(X_J)\} \leq e^{10C/9}. \quad (6.4)$$

Thus full OVERLAP is already the $q = 10/9$ Nelson theorem in variational coordinates. The proof order must be

$$\boxed{\begin{array}{c} \text{complete temporal packets and controlled-shell assembly} \\ \Downarrow \\ \text{full OVERLAP} \implies \text{R-087 CORE} \implies \text{Nelson} \end{array}} \quad (6.5)$$

Placing R-066 one-use after full OVERLAP would count a necessary input as a later consequence. REG and separately named packets remain useful sufficient architecture, but they are not logically necessary labels for the equivalence (6.2). This correction is registered as [AUDIT-2026-07-25-A13-OVERLAP-NELSON-CHAIN](#).

7 Attempt and evidence map

1. **Global polar/Douglas compression – proved.** Arbitrary progressive revisit has no terminal CM multiplicity loss; (2.2)–(2.10).
2. **R-088 quartic terminal bridge – extended.** The pure-control quartic payload is form-payable for the full cylindrical-simple progressive class.
3. **Complete cross- k Cartan sum – proved reduction.** The exact first-order operator, Fourier trace, and two coefficient tails are (3.4)–(3.11).
4. **Nonlinear coefficient energy – open.** The exact remaining estimate is (3.12); lower-order b, c tails are not paid.
5. $s = 1/4$ **quartic interpolation – proved.** The powers are $5/16, 9/16$, the slack is $1/8$, the model moment is eight, and the gap is $2^{-C/2}$.
6. **Pure homogeneous quartic closure – ruled out.** The exact harmonic $h_{32} > 0$ forces squared response c^2 against quartic payload c^4 .
7. **Rational Taylor coordinates – proved.** The conditional identity is (5.3) and the spectral criterion is (5.4).
8. **Eta repairs the production mean – ruled out.** The actual scalar ray has $L/e = -1/432$ for every eta.

9. **Centered same-root rational packet alone – ruled out.** The covariance-matched adapted fixture (5.10) is strictly negative; the complete forest is needed.
10. **OVERLAP then one-use – corrected.** Full OVERLAP is already equivalent to Nelson; packet assembly belongs before it.

The next analytic target, R-090, should attack (3.12) and (5.11) as a coupled coefficient/forest problem. The successful terminal compression should be treated as an available lemma, not reproved shell by shell.

8 Devil’s-advocate review

1. **Objection:** (2.2) silently assumes disjoint time ranges or a one-shot control. **DISMISSED.** The identity $C_J = T_J T_J^*$ and polar decomposition use the complete operator; overlap and repetition are inside both sides. A repeated one-dimensional range saturates, rather than violates, the contraction.
2. **Objection:** martingale orthogonality in (2.7) is scalar only. **DISMISSED.** Conditional expectation is an orthogonal projection on Hilbert-valued L^2 . At finite cutoff take the target Hilbert space to be H_{CM} . The deterministic H^2 multiplier may be applied before the same argument.
3. **Objection:** (3.7) loses cross terms between physical shells. **DISMISSED.** Those shells are summed exactly into b and c in (3.2)–(3.4) before squaring. Only the elementary split between the value and derivative coefficient is estimated in (3.8).
4. **Objection:** the two-tail bound proves controlled Cartan CFAR. **VALID WITH BOUNDARY.** It proves the exact reduction (3.10). The nonlinear coefficient energy (3.12) is unproved, so CFAR and OVERLAP remain open.
5. **Objection:** the h_{32} fixture refutes every possible quartic Cartan estimate. **DISMISSED.** It refutes only a homogeneous pure-quartic implication. Additive form constants, model moments, and the lower-order coefficient channels are outside the no-go.
6. **Objection:** the rational fixture is not centered. **DISMISSED.** The fixture (5.9) uses $G \sim N(0, 1)$ with $\mathbb{E}G = 0$ and $V = \Gamma = 1$. Its failure is same-root coefficient alignment, not a missing first-moment centering assumption.
7. **Objection:** (5.10) disproves the global shifted-Hessian form. **DISMISSED.** It tests the Taylor/Wick subpacket with its coefficient but omits the complete backward-heat/lower-chaos forest. It is a method no-go and identifies the required residual (5.11), not a global counterexample.
8. **Objection:** CORE plus Nelson still needs R-066 afterward. **DISMISSED.** Equation (6.2) is an equality of infimum and logarithmic moment. R-066 is an input to proving the uniform lower bound (6.3), not an extra implication after it.

9 Executable verification and no-overclaim statement

The primary executable verifies the exact rational arithmetic, the harmonic coefficient, the Fourier trace fixture, subcritical exponent ledger, and finite-dimensional compression fixtures. The independent executable shares no imports with it and reconstructs the polar contraction by SVD, the martingale ledger by exact enumeration, the Fourier trace from an explicit operator matrix, the harmonic by high-precision quadrature, and the rational identities from independent random matrices. Run

```
cd codes/foundations
python a13_classii_progressive_covariance_compression_rational_mean_spectral_boundary_verify.py
```


The manifest also records the equivalent one-line command from the repository root.

R-089 establishes global progressive terminal covariance compression and its martingale/spatial ledgers, the resulting general-progressive pure-control quartic terminal bridge, the exact complete-cross-shell Cartan Fourier trace reduction, the $s = 1/4$ subcritical quartic ledger, the exact pure-homogeneous-quartic no-go, the rational Taylor-coordinate criterion and production mean-spectral obstruction, and the OVERLAP–Nelson chain correction. It does not establish (3.12), the complete rational residual bound (5.11), full REG, uniform OVERLAP, Nelson, an interacting measure, or Sector-A closure. No tier promotion is made.

Result footer

Result ID: A13-CLASSII-PROGRESSIVE-COVARIANCE-COMPRESSION-
RATIONAL-MEAN-SPECTRAL-BOUNDARY

Ledger: R-089.

Precise statement: Global polar/Douglas compression controls every finite-cutoff cylindrical-simple progressive terminal CM coordinate without partition, overlap, or revisit loss. Hilbert martingale orthogonality gives weighted terminal one-use. The complete cross-shell Cartan derivative has an exact first-order Fourier trace and two-tail reduction. The rational packet has an exact Taylor-coordinate mean-spectral criterion. Full OVERLAP is equivalent through R-087 CORE to $q=10/9$ Nelson.

Scope: Fixed finite cutoff and positive floor, under the hash-pinned A8, R-087, and R-088 conventions. The Cartan result reduces to coefficient energy (3.12); the rational forest residual (5.11) is open.

Dependencies: A8; R-076; R-081; R-084; R-087; R-088.

Evidence grade: ANALYTIC, EXACT, EXECUTED.

Reproduction command: python codes/foundations/a13_classii_progressive_covariance_compression_rational_mean_spectral_boundary_verify.py

Expected output: Primary 56/56; independent 44/44; integrated 169/169; aggregate 269/269 PASS; exit zero.

Audits: AUDIT-2026-07-25-A13-R088-PROGRESSIVE-TERMINAL-CM-BRIDGE; AUDIT-2026-07-25-A13-OVERLAP-NELSON-CHAIN.

Negative results: NG-2026-07-25-A13-PURE-QUARTIC-CARTAN-HOMOGENEITY; NG-2026-07-25-A13-RATIONAL-ETA-MEAN-SPECTRAL-CLOSURE. Both are method no-gos.

Falsification gate: Failure of global compression, martingale ledger, Cartan Fourier trace, harmonic, rational identity, production sign, CORE equivalence, evidence hash, PDF contract, or assertion count.

Tier before / after: T4 / T4; no promotion.

No-overclaim statement: No nonlinear Cartan coefficient-tail energy, complete same-root rational heat/forest packet, REG, uniform OVERLAP, Nelson, interacting measure, removal limit, or Sector-A closure.

Next required action: Prove coefficient energy (3.12) and complete rational residual (5.11), then assemble them, R-066 one-use, and all temporal packets inside full OVERLAP.